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GroundOne: A Specialist System One for Groundedness Verification in Local Retrieval-Augmented Generation

Fine-tuning compact language models as single-pass, calibrated verifiers on Apple Silicon

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Summary

Retrieval-Augmented Generation (RAG) is the standard way to give language models access to private documents, and running the whole pipeline on the user's own device is now practical on consumer hardware, which helps with privacy, offline use and cost. Retrieval does not, however, eliminate hallucination: generated answers still contain claims that the retrieved context does not support. Detecting such unfaithful claims as the answer is produced is a prerequisite for trustworthy local assistants. Since the check must run on every answer, it has to fit on the same constrained machine as the generator.

Automatic faithfulness verification currently follows two main paradigms. *Encoder-based checkers* (NLI and fact-verification models such as AlignScore, MiniCheck and HHEM) are small and fast, but they are fixed-function classifiers: their context window is limited and they cannot take instructions or policies expressed in natural language. *LLM-as-a-judge* approaches (e.g. RAGAS, ARES, Lynx) are far more flexible, but they are autoregressive. The verdict is generated token by token, so latency and cost grow with the length of the output. Small models often fail to respect the output format, and the confidence they express is rarely well calibrated. A third family has recently emerged: **System One models** (the proprietary Jev and the open re-implementations SemIf and Rizzo-Flow). These answer closed-form questions in a single forward pass, reading the probability of each answer option directly from the logits instead of generating text. The evidence for their advantage is mostly vendor-reported or small-scale, and the open implementations are general-purpose.

Three questions therefore remain open. The first is whether a **task-specialised, fine-tuned compact System One** can match the accuracy of larger generative judges and of dedicated verification encoders on a real task. The second is how much of its advantage comes from specialisation, how much from the single-pass readout, and how much from evaluating several claims over one shared context. The third is whether it delivers probabilities that are usable because they are calibrated. All three have to be answered **under the memory and latency constraints of consumer hardware** (16 GB of unified memory). No independent, controlled study appears to address them. In particular, the fairest control, a same-size LLM

that emits a single answer token, is rarely included in comparisons.

This thesis designs, trains and evaluates **GroundOne**, a System One specialised in *one* decision: for each sentence of an answer, whether the retrieved context supports it, contradicts it or does not mention it. Compact open models (Qwen2.5 0.5B and 3B) are fine-tuned with LoRA on human-labelled answers of RAGTruth, with a training loss restricted to the logits of the three answer options. At inference, the context and the answer are encoded **once**, and the decisions for all K sentences of the answer are then read **in parallel**, without generating text and without parsing any output. Each decision yields three probabilities, from which everything else follows: a binary groundedness score, and a strict or a lenient notion of error that the user can choose without retraining, whereas binary encoders such as MiniCheck and HHEM provide a single score. A post-hoc calibration is applied on top. The model is implemented on MLX and runs entirely on a 16 GB Apple Silicon laptop, where it is integrated in an end-to-end local RAG demo that flags unsupported sentences in real time.

The system is evaluated on three human-labelled test sets that are built and frozen before any training, with strict separation from the training data and automated leakage checks: the RAGTruth test split, for the three-way decision and for results as a function of K; LLM-AggreFact, for the binary task and comparison with published systems; and VitaminC, a balanced test of the distinction between contradicted and unmentioned content whose labels have a known inter-annotator agreement. To isolate each source of gain, GroundOne is compared against a ladder of baselines: (1) generative LLM judges of the same size, in a terse and a verbose variant, together with published results of larger models; (2) the same models with single-token readout but without fine-tuning; (3) a general-purpose System One (Rizzo-Flow); (4) specialised verification encoders (MiniCheck and HHEM); and (5) simple cues that never verify the sentence against the document, namely word overlap and the wording of the sentence alone. Latency is measured as a function of the number of decisions per context. The metrics cover discrimination (balanced accuracy, macro recall, AUROC), calibration (ECE, Brier score, selective-prediction curves), latency (p50/p90/p99 per answer, throughput), peak memory and output-format failure rate. All claims are stated as falsifiable hypotheses, and negative results will be reported.

Acknowledgements

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Table of Contents

List of Figures VIII

1 Introduction 1
- 1.1 Motivation . . . 1
- 1.2 System One models . . . 1
- 1.3 Research questions . . . 1
- 1.4 Contributions . . . 1
- 1.5 Thesis structure . . . 1

2 Background and Related Work 2
- 2.1 Transformer inference: prefill, decode and the KV cache . . . 2
- 2.2 System One models . . . 2
- 2.3 Retrieval-augmented generation and hallucination . . . 2
- 2.4 Groundedness verification . . . 2
- 2.5 Calibration and selective prediction . . . 2
- 2.6 Parameter-efficient fine-tuning . . . 2
- 2.7 Research gap . . . 2

3 GroundOne: Method and Implementation 3
- 3.1 Task and decision . . . 3
- 3.2 Reading all verdicts in one forward pass . . . 4
- 3.3 Training objective . . . 4
- 3.4 Calibration . . . 5
- 3.5 Software . . . 6

4 Data 7
- 4.1 Task and labels . . . 7
- 4.2 Source datasets . . . 8
- 4.3 Construction . . . 9
- 4.4 The resulting dataset . . . 10
- 4.5 Reliability of the test sets . . . 11

4.6 Exploratory analysis 13
4.7 Leakage control 14

5 Experimental Setup 16
5.1 Models 16
5.2 Baselines 16
5.3 Metrics 18
5.4 Latency protocol and hardware 18

6 Results 19
6.1 Training 19
6.2 Accuracy on the RAG test set (T1) 19
6.3 Comparison with the baselines 21
6.4 Transfer to a different domain (T3) 22
6.5 Calibration and selective prediction 22
6.6 Latency as a function of K 24
6.7 Error analysis 25

7 Discussion, Limitations and Threats to Validity 26
7.1 When a specialist System One pays off 26
7.2 Limitations 26
7.3 Threats to validity 26

8 Conclusion and Future Work 27
8.1 Summary of findings 27
8.2 Future work 27

Bibliography 28

List of Figures

4.1 Class balance of the labelled sentences in each split. RAGTruth (train, development and T1) is about 88–91 % supported; T3 is sampled to 1,000 pairs per class. 11

4.2 Distribution of K, the number of scored sentences per answer, in training and in T1. Dashed lines mark the bands used to report results as a function of K. 12

4.3 Share of contradicted or not-mentioned sentences by the model that wrote the answer, in training and in T1. 13

4.4 Word overlap between a sentence and its document, by class, in training, T1 and T3 (mean in brackets). 14

4.5 Share of contradicted or not-mentioned sentences by position in the answer. 15

6.1 Three-way balanced accuracy on the development set during training of the 0.5B model, overall (left) and per task (right). The dotted line marks chance. The 3B shows the same shape on its 200-answer development subset (`figures/training-g1-3b-b1`). 20

6.2 T1: AUROC of P(supported) for every system (left), and three-way balanced accuracy for the systems with a three-way output (right). Rizzo-Flow is evaluated on 300 answers and the untrained 3B on 600. 21

6.3 Reliability of P(supported) of the 0.5B before (raw) and after calibration on the development set, for the development set, T1 and T3. Points are bins with at least 20 sentences. 23

6.4 Risk–coverage curves of the binary decision of the 0.5B with calibrated probabilities: error rate as the least confident sentences are deferred. 23

6.5 Median time per answer as a function of K on the same 200 T1 answers (in-process MLX, one answer at a time, M4 with 16 GB), for GroundOne, the System One with one question per sentence, and the terse and verbose generative judges. Logarithmic scale. . . 24

6.6 On the latency subset: three-way balanced accuracy (format failures counted as errors) and share of sentences whose verdict could not be parsed. 25

Chapter 1

Introduction

1.1 Motivation

1.2 System One models

1.3 Research questions

1.4 Contributions

1.5 Thesis structure

Chapter 2

Background and Related Work

2.1 Transformer inference: prefill, decode and the KV cache

2.2 System One models

2.3 Retrieval-augmented generation and hallucination

2.4 Groundedness verification

2.5 Calibration and selective prediction

2.6 Parameter-efficient fine-tuning

2.7 Research gap

Chapter 3

GroundOne: Method and Implementation

GroundOne is a compact language model used as a System One: it reads a document and an answer once and returns a verdict for every sentence of the answer by reading the logits of a few answer letters, without generating text. This chapter describes the decision it makes, the prompt that carries all the decisions of one answer in a single forward pass, the training objective, the calibration of the output probabilities and the software that implements them.

3.1 Task and decision

For every sentence of an answer GroundOne makes one decision among three classes, defined in Section 4.1: the sentence is *supported*, *contradicted* or *not mentioned* with respect to the document. The three classes are mapped to the letters A, B and C, and the model outputs a probability for each of them. A binary groundedness score and the strict and lenient policies are not separate questions to the model; they are computed from these three probabilities, so that a single forward pass serves every use. An earlier design asked the model four different questions about each sentence, including the two policies written out in the prompt. Those questions were deterministic functions of the three-way class, so they inflated the number of decisions artificially and gave the model conflicting targets for the same sentence. The single three-way decision replaces them.

3.2 Reading all verdicts in one forward pass

The prompt contains a short instruction, the document and the answer. Every sentence of the answer is numbered, and every sentence that carries a decision is followed by a marker, the string " `Verdict:`". Sentences without checkable content (Section 4.3) are kept as context but receive no marker. The model processes the whole sequence once, and at the last token of each marker the logits of the letters A, B and C are read and normalised with a softmax restricted to these three letters. Because a decoder only attends to earlier positions, the verdict on sentence i depends on the document and on sentences $1, \ldots, i$, which is what verification needs: references such as pronouns point backwards. Nothing after the last marker affects any verdict, so the sequence ends there.

Two details make the readout efficient and exact. First, the prompt is built by tokenising its pieces separately and concatenating them, so the position of every marker is known exactly and is identical at training and inference time. Second, the output projection onto the vocabulary is applied only at the marker positions. The transformer body runs over the full sequence, but the large vocabulary matrix is multiplied only by the few hidden states that are read.

This design determines the cost of a verification. If S is the length of the document and answer, K the number of decisions and m the length of a marker in tokens, the System One costs one prefill of $S + Km$ tokens. A generative judge must also decode its verdicts: with n tokens per verdict and a per-token decoding time t_{dec}, its cost is roughly $\mathrm{prefill}(S) + K\,n\,t_{\mathrm{dec}}$. Since m is three tokens and decoding is much slower per token than prefill on the target hardware, the gap widens as K grows. An alternative design, used in earlier experiments and kept as a fallback, reads the shared prefix once and then appends one short question per sentence, scored in a batch; it needs a key-value cache replicated K times, whereas the marker design needs a single pass.

3.3 Training objective

GroundOne is obtained from Qwen2.5-0.5B-Instruct and Qwen2.5-3B-Instruct [1] with low-rank adaptation (LoRA) [2]. Adapters of rank 16, with scaling factor 2 and dropout 0.05, are attached to all linear layers of all transformer blocks, since covering every layer matters more than a high rank [3]. The base weights stay frozen in 8-bit precision. One training example is one answer with all its scored sentences, and the loss is computed at every marker of the answer.

The loss is a cross-entropy over the three letters only, so the prompt itself is never supervised and the training signal is entirely the verdicts. Because about 88 % of the training sentences are supported, the classes are weighted so that each

contributes equally to the loss: with the training distribution this gives weights of about 0.38 for supported, 7.5 for contradicted and 4.4 for not-mentioned sentences. Without such weights a model can reach a low loss by answering "supported" almost always. A useful consequence is that a model which learned only the class frequencies would stay at a loss of $\ln 3 \approx 1.10$, so a loss below this value shows that the text is being used.

Optimisation uses AdamW without weight decay, a learning rate of 10^{-4} reached after a linear warm-up over the first 3 % of the steps and then decayed to zero with a cosine schedule, and one epoch over the training answers. Answers of similar length are grouped into micro-batches under a budget of padded tokens, and gradients are accumulated so that each optimiser step sees about 16 answers, or about 110 decisions; this keeps a few contradicted sentences in every step despite their rarity. Gradient checkpointing keeps memory within the 16 GB of the laptop. The model is evaluated on the development set at regular intervals, and the checkpoint with the highest three-way balanced accuracy, averaged over the three tasks, is kept. Section 5.1 lists the settings of each run.

3.4 Calibration

The class weights train the model as if the three classes were equally frequent, so its raw probabilities are not calibrated to the real frequency of errors. A post-hoc map is therefore fitted on the development set: the three logits are divided by a temperature T and shifted by a bias per class, $\mathrm{softmax}(z/T + b)$ with the bias of the supported class fixed at zero, and T and b minimise the negative log-likelihood of the development labels [4]. A map of this kind changes the probabilities but not their ranking, so measures such as AUROC are unaffected. The bias, however, can change which class is most probable, which matters for how decisions and probabilities are used (Section 6.5).

Calibration is measured mainly with the expected calibration error (ECE) [5, 4]. The predictions are divided into M bins of equal width according to their confidence, with $M = 15$ in this thesis. For each bin B_m the mean confidence $\mathrm{conf}(B_m)$ is compared with the observed frequency $\mathrm{acc}(B_m)$, and the gaps are averaged with weights proportional to the size of the bins:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \Big|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\Big|. \tag{3.1}$$

In the binary variant the confidence is $P(\text{supported})$ and the observed frequency is the share of supported sentences in the bin; in the top-label variant for three classes the confidence is the highest probability and the observed frequency is the share of correct predictions. ECE does not compare the class distribution of the

predictions with that of the test set; it asks whether, among the sentences to which the model assigns a given confidence, that confidence matches reality. It depends on the choice of bins, and a constant predictor that always outputs the base rate has an ECE close to zero while discriminating nothing. For these reasons ECE is always reported together with the Brier score, the negative log-likelihood and reliability diagrams.

3.5 Software

GroundOne is implemented as an installable Python package on MLX [6, 7], the array framework for Apple silicon, and every training and inference step described above runs in it on the same laptop. The package contains the dataset builder and its checks, the training loop, the calibration, the evaluation on the frozen test sets, the baselines and a memory guard that runs long jobs one at a time and stops a job before the system runs out of memory. A command-line interface and an HTTP server expose the verifier; the server offers a native endpoint and an endpoint compatible with the interface used by other System One models, so that existing clients can use GroundOne by changing the address. All model computation runs on a single long-lived worker thread, because the Metal context of the GPU must not move between threads.

Chapter 4

Data

GroundOne learns a single decision, and the data must teach exactly that decision and measure it reliably. This chapter first defines the decision and its labels, then describes the three public datasets that provide them and the way they are turned into training, development and test sets. The last sections report how the reliability of the test sets was checked without new annotation, what an exploratory analysis of the result reveals, and how leakage between training and test data was ruled out.

4.1 Task and labels

The input is a retrieved document and an answer written by a language model, split into sentences. For every sentence the model decides whether, *with respect to the document*, the sentence is *supported*, *contradicted* or *not mentioned*. A sentence is supported when everything it states is said or directly implied by the document. It is contradicted when some part of it conflicts with the document, and not mentioned when it adds information that the document neither contains nor contradicts. A sentence that contains both a conflicting and an unverifiable part is labelled as contradicted, because a contradiction is the more serious error. Only the document counts: a statement that happens to be true in the world but is absent from the document is still not mentioned.

The number of decisions for one answer is the number of its scored sentences, denoted K. This quantity is central to the thesis. A System One reads the document and the answer once, so each additional decision is almost free, while a generative judge has to produce text for every sentence. The comparison with generative judges is therefore made as a function of K, and the data must contain long answers as well as short ones.

The model outputs three probabilities per sentence, and every other quantity is derived from them rather than learned separately. The binary groundedness score,

used for comparison with published binary benchmarks, is $P(\text{supported})$. A *strict* policy, under which anything not stated in the document is an error, flags a sentence when $P(\text{supported}) < \tau$. A *lenient* policy, under which only contradictions are errors, flags it when $P(\text{contradicted}) > \tau$. The user can thus choose the notion of error at inference time without retraining the model.

4.2 Source datasets

The dataset is deliberately built from as few sources as possible: one for training and three for testing (Table 4.1). Fewer sources mean a single definition of the labels, a single text format and fewer places where the construction can go wrong.

Table 4.1: Source datasets and their role in GroundOne.

Dataset	Licence	Role	Labels
RAGTruth [8]	MIT	train, dev, test T1	character spans of four hallucination types on LLM answers
LLM-AggreFact [9, 10]	CC BY-ND 4.0	test T2 only	binary (supported / unsupported) per claim
VitaminC [11]	CC BY-SA 3.0	test T3 only	supports / refutes / not enough information

RAGTruth. RAGTruth [8] contains 17,790 answers that six language models (GPT-3.5, GPT-4, Llama-2 7B, 13B and 70B, and Mistral-7B) wrote in three retrieval-augmented tasks. In question answering the context is a set of passages retrieved for an MS MARCO question [12]. In summarisation it is a news article from CNN/DailyMail [13, 14] or from recent news, and in data-to-text generation it is a Yelp business record in JSON [15]. Two trained annotators marked the character spans of each answer that the context does not support, with a third review in case of disagreement, and assigned each span one of four types: evident or subtle *conflict*, and evident or subtle *baseless information*. The paper reports an agreement of 91.8 % at the level of answers and 78.8 % at the level of spans. It is the only public dataset that combines natural multi-sentence answers from several models with a distinction between contradicted and unsupported content, which is why it is the only training source. Its official split separates 450 test sources from the remaining training sources.

LLM-AggreFact. LLM-AggreFact [9, 10] aggregates eleven human-labelled fact-checking datasets of (document, claim) pairs. They cover summarisation

(AggreFact-CNN and XSum [16], TofuEval [17]), retrieval-augmented and long-form question answering (RAGTruth [8], ClaimVerify [18], ExpertQA [19], LFQA [20]) and fact checking (WiCE [21], REVEAL [22], FactCheck-GPT [23]). Its public leaderboard makes results comparable with published systems. The licence states that the data must not be used to train models [24], so in this thesis it is used only for evaluation: nothing is trained, calibrated or selected on any of its splits.

VitaminC. VitaminC [11, 25] is a fact-verification dataset built from Wikipedia revisions that change a fact. For each revision, trained annotators wrote claims that one version of the sentence supports and the other refutes; when no contradiction could be written, the pair was labelled as having not enough information. Because the two versions of the evidence are almost identical, a model must look at the exact fact rather than at word overlap. On a re-annotated sample of 2,000 pairs the authors report a Fleiss' κ of 0.71. VitaminC plays a specific role in this thesis. RAGTruth publishes no agreement for the hallucination *type*, so it cannot alone show that the boundary between contradicted and not mentioned is learned. VitaminC provides that boundary with labels of known reliability, in a different domain.

Several other candidates were examined and not adopted. Natural language inference corpora such as ANLI [26] have short premises far from retrieval-augmented contexts, and VitaminC covers the same three-way distinction with factual text. Synthetic or binary-only training data, such as the data released with MiniCheck [9], does not carry the three-way labels. Datasets whose labels were generated by a language model, or which reuse RAGTruth test answers, were excluded to keep the test sets independent and the labels human.

4.3 Construction

Sentence segmentation. Answers are first split at line breaks, because they often contain lists and headings, and each line is then segmented with PySBD [27], a rule-based segmenter that handles abbreviations and decimal numbers. Leading list markers such as "3." or "*" are removed, since they are layout rather than content. The same segmenter is used for training, evaluation and inference, so the model always sees sentences cut in the same way.

Sentences without checkable content. Some sentences make no claim about the document: greetings, headings that introduce a list, citation-only lines such as "(Passage 2)" and refusal formulas. They remain in the numbered answer as context but are not decisions, and they carry no label. A short rule detects them, and it applies to 8.2 % of all RAGTruth sentences, mostly list headings. The

rule was written in advance and adjusted twice, only for formatting: list markers and citation-only lines were handled after the comparison with LLM-AggreFact described in Section 4.5 showed them as the main source of mismatches, and a few conversational formulas found in the training data were added. No change was made to the labels.

Labels. A scored sentence is labelled as contradicted if it overlaps a conflict span, as not mentioned if it overlaps only baseless-information spans, and as supported otherwise. Only 27 of the 14,226 annotated spans (0.19 %) touch no scored sentence, so almost no annotation is lost in the conversion. For LLM-AggreFact the binary label is kept as released. VitaminC labels are mapped directly: *supports* to supported, *refutes* to contradicted and *not enough information* to not mentioned.

Exclusions and splits. Answers flagged by RAGTruth as incorrect refusals or truncated are removed, as are answers without any scored sentence and answers whose document and answer together exceed 2,048 tokens, the length limit imposed by memory on the 16 GB laptop. The official RAGTruth test split becomes test set T1. The training sources are shuffled with a fixed seed: 10 % of them form the development set and the rest the training set, so that answers about the same source never appear in both. Six training answers about one news article that also appears in the LLM-AggreFact test set are removed. From the real part of the VitaminC test split, which is unbalanced, 1,000 pairs per class are sampled with a fixed seed to form T3.

4.4 The resulting dataset

Table 4.2 summarises the five files. The training set contains 13,203 answers and 90,795 labelled sentences. Records are stored per answer, with one label per sentence, so the same data can feed different ways of reading the K decisions of an answer.

Table 4.2: The GroundOne dataset. A decision is a labelled sentence; T2 is binary.

Split	Source	Records	Decisions	Supported	Contradicted	Not mentioned
Train	RAGTruth	13,203	90,795	79,898	4,039	6,858
Dev	RAGTruth	1,468	9,996	8,827	410	759
T1	RAGTruth test	2,647	17,284	15,745	584	955
T2	LLM-AggreFact test	29,320	29,320	22,832	6,488 unsupported	
T3	VitaminC test	3,000	3,000	1,000	1,000	1,000

RAGTruth is strongly imbalanced: about 88 % of the labelled sentences are supported, and contradictions are only 3 to 4 % (Figure 4.1). This is the natural rate of errors in model answers, and it is kept: the test set reflects what a verifier meets in practice, while the imbalance is handled during training. T3 compensates with a balanced three-way test. Answers have 6.9 scored sentences on average in training and 6.5 in T1, with a maximum of 28 (Figure 4.2). For the analysis by length, T1 is divided into four bands of K: 756 answers with 1–4 sentences, 1,288 with 5–8, 497 with 9–13 and 106 with 14 or more. Cells with fewer than 30 sentences of a minority class, such as contradicted sentences in question answering, are reported but marked as indicative.

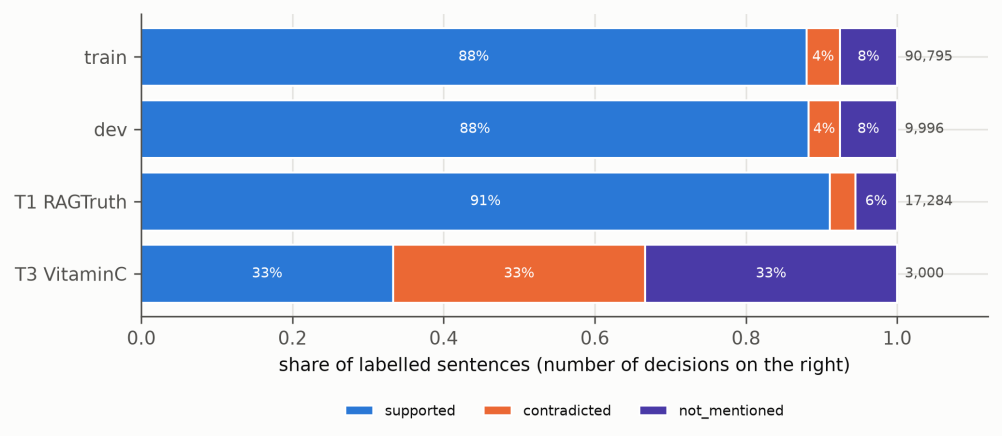


Figure 4.1: Class balance of the labelled sentences in each split. RAGTruth (train, development and T1) is about 88–91 % supported; T3 is sampled to 1,000 pairs per class.

4.5 Reliability of the test sets

The test sets were checked without new annotation, using evidence that is independent of the construction described above (Table 4.3). All test labels come from the human annotators of the published datasets. The two choices that could introduce errors, segmentation and the conversion from spans to sentence labels, were compared with LLM-AggreFact, whose authors converted the same RAGTruth test answers into sentence-level claims independently. The segmenter recovers 96.3 % of their claims exactly, and only 8.5 % of the scored sentences are absent from their set, mostly sentences that their undocumented filter of “non check-worthy” claims removed. On the 15,492 sentences present in both, the binary labels agree in 98.75 % of cases, and all 193 disagreements go in the same direction: the span rule marks a sentence as unsupported where LLM-AggreFact considers it supported. Finally, an

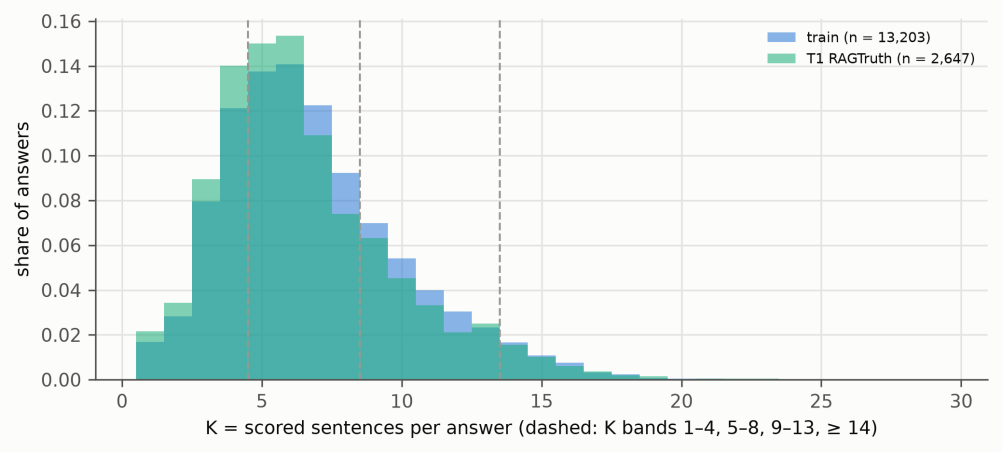


Figure 4.2: Distribution of K, the number of scored sentences per answer, in training and in T1. Dashed lines mark the bands used to report results as a function of K.

encoder baseline evaluated on the unchanged T2 reproduces its published score (74.0 against 73.5), which validates the evaluation code.

Table 4.3: Reliability checks run before the test sets were frozen.

Check	Criterion	Result
Label provenance	every test label comes from human annotators	pass
Segmentation	$\geq 90\,\%$ of LLM-AggreFact claims recovered, $\leq 10\,\%$ of sentences absent from it	96.3 % / 8.5 %
Label derivation	$\geq 97\,\%$ agreement with LLM-AggreFact's binary labels	98.75 %
Reference baseline	MiniCheck-RoBERTa on T2 within 1.5 points of its published score	74.0 vs 73.5
Leakage	no shared source, document or (document, sentence) pair between training and test	pass

These checks validate the binary side of T1. The distinction between contradicted and not mentioned rests on the RAGTruth annotators, whose agreement on the hallucination type is not published. Moreover, a recent re-annotation of the RAGTruth test split found 1.68 times more hallucinated answers than the original labels [28], which suggests that T1 under-counts the two minority classes. For these reasons the three-way results on T1 are always read together with those on T3. After these checks the three test files are frozen with cryptographic hashes, and the dataset builder refuses to overwrite them with different content.

4.6 Exploratory analysis

An exploratory analysis of the sources and of the built dataset revealed four properties that shape how the results must be read.

Errors depend on the task and on the model. Data-to-text answers are hallucinated two to four times as often as answers in the other tasks, and answers written by Llama-2 and Mistral contain most of the hallucinations, while GPT-3.5 and GPT-4 answers are almost always faithful (Figure 4.3). Results are therefore reported per task, since a single average would mostly reflect data-to-text.

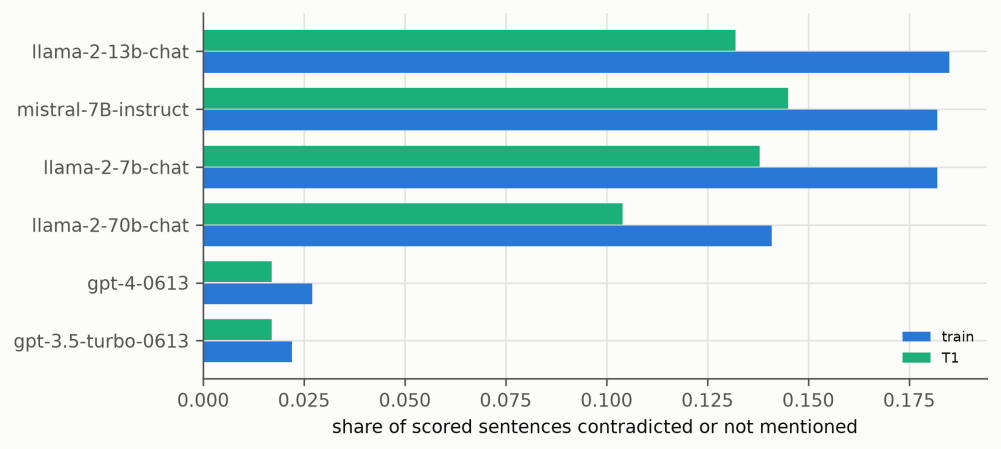


Figure 4.3: Share of contradicted or not-mentioned sentences by the model that wrote the answer, in training and in T1.

The test answers are cleaner than the training answers. In every task, and for every generator, T1 has a lower share of contradicted and not-mentioned sentences than the training set; in question answering, for instance, 0.6 % of T1 sentences are contradicted against 1.6 % in training. Metrics that depend on the base rate, such as precision or F1 on the minority classes, are therefore not comparable between development and test. The primary metrics are balanced accuracy, macro recall and AUROC, which do not depend on it.

Contradictions are mostly evident. About 85 % of the contradicted training sentences come from spans annotated as *evident* conflicts, such as wrong numbers or names, and only 147 from *subtle* conflicts. The contradicted class is therefore mostly blatant contradictions, and subtle contradictions cannot be evaluated separately.

Surface cues exist and define the baselines. Two simple cues predict the labels without verifying the sentence against the document. The first is word overlap. The share of a sentence's content words that also appear in the document is higher for supported than for contradicted and not-mentioned sentences, in all three test sets (Figure 4.4). On its own it separates supported from unsupported sentences with an AUROC of 0.74 on T1 and 0.70 on T3. The second cue is the wording of the sentence. A classifier that sees only the sentence reaches a three-way balanced accuracy of 0.66 on data-to-text, because some details, such as the number of reviews or the availability of Wi-Fi or valet parking, are often invented in Yelp descriptions. In question answering and summarisation the same classifier stays at chance (0.36 and 0.35), and on T3 as well (0.33), where the contrastive design removes the cue. A verifier is useful only to the extent that it beats these cues, so both are reported as baselines, per task.

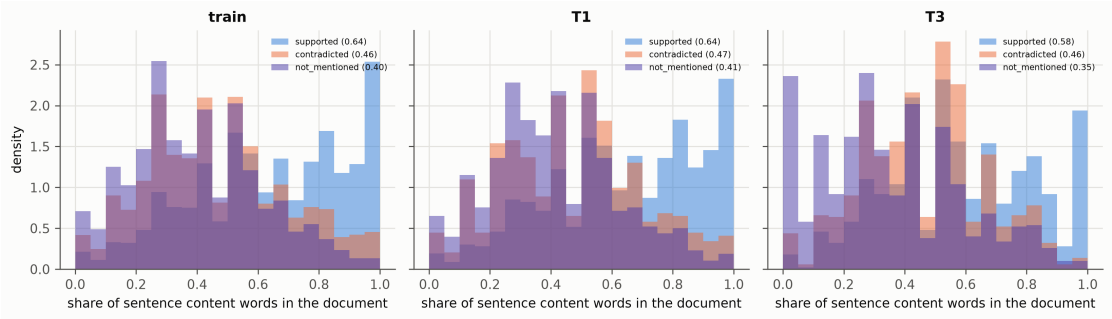


Figure 4.4: Word overlap between a sentence and its document, by class, in training, T1 and T3 (mean in brackets).

Finally, unsupported sentences are not concentrated at the end of an answer. The first sentence is rarely unsupported, the share peaks at positions three to five and stays between 8 and 11 % in later positions (Figure 4.5). Long answers still contain errors near their end, so accuracy at high K matters.

4.7 Leakage control

Splits are made by source, never by sentence, and an automatic check runs on every build. It fails if a source, a document or a (document, sentence) pair of the training or development set appears in any test set, and if the development and training sets share a source. The check passes. It also reports sentence strings that occur in both training and test with different documents: these are 788 template sentences, such as "The business has a rating of 4.0 stars.", which are expected in data-to-text and do not reveal any test label. Whether the public datasets used

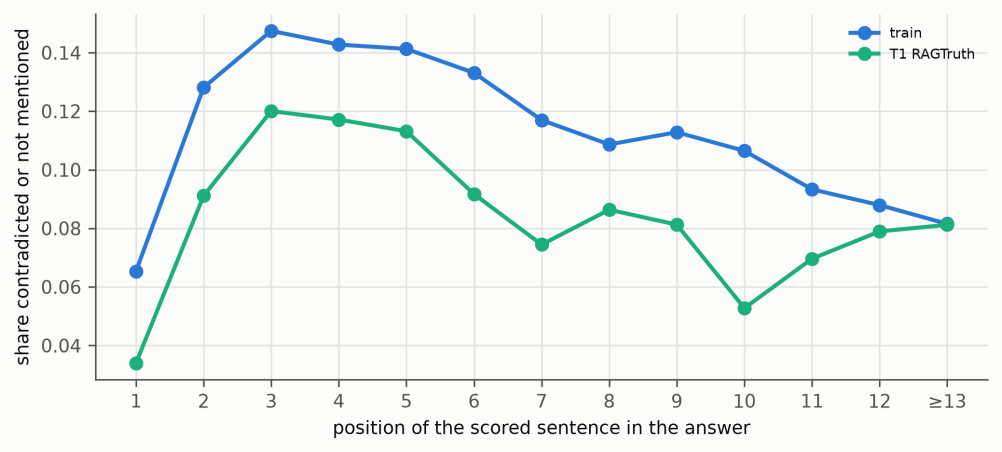


Figure 4.5: Share of contradicted or not-mentioned sentences by position in the answer.

here were part of the pre-training data of Qwen2.5 cannot be verified, and this remains a limitation shared by most work on public benchmarks.

Chapter 5

Experimental Setup

This chapter describes the models that were trained, the systems they are compared with, the metrics and the measurement protocol. All experiments run on one machine, a MacBook Pro with an Apple M4 chip and 16 GB of unified memory, so that every result reflects what a verifier can do on the device where a local assistant runs.

5.1 Models

Both GroundOne models start from the instruction-tuned Qwen2.5 checkpoints [1] converted to MLX with 8-bit weights, and both are trained with the recipe of Section 3.3. They differ only in size, in the number of training answers and in how micro-batches are formed, which memory dictates. The 0.5B model is trained on all 13,203 training answers: one epoch of 802 optimiser steps took 5.4 hours, with an evaluation on the full development set every 150 steps. The 3B model is about six times slower per token, so it is trained for one night on a random subset of 6,500 answers (about 45,000 decisions); its development evaluation uses a fixed subset of 200 answers, because a full pass would take about an hour. In both cases the checkpoint with the best development score is kept and then calibrated as described in Section 3.4.

5.2 Baselines

The baselines isolate the sources of any advantage. Each one answers a specific question.

Cues that never read the document as a whole. Two simple predictors give the floor that any verifier must clear. The first scores a sentence by the share of

its content words that also occur in the document. The second is a bag-of-words classifier trained on the training sentences alone, without the document. Both are described in Section 4.6.

Untrained readout (B1). The same Qwen2.5 models, without fine-tuning, are asked the three-way question for each sentence and their answer-letter logits are read. Because an untrained model has no reason to treat the `Verdict:` marker as a question, this baseline uses the alternative format with one explicit question per sentence (Section 3.2). It measures what the System One mechanism gives before any specialisation.

Generative judges (B0a, B0b). The untrained models are also used in the way that is common in practice: they receive the document, the numbered answer and the list of sentences to check, and they write a verdict for each sentence, with greedy decoding. The *terse* judge writes one line "`<number>: <letter>`" per sentence; the *verbose* judge writes one short sentence of reasoning followed by the verdict. A verdict that cannot be parsed is a format failure; it is counted as an error in accuracy and reported separately. These judges are the main reference for latency.

A general-purpose System One (B3). Rizzo-Flow [29] is an open re-implementation of the System One idea with a 4B model fine-tuned on generic typed decisions. It is queried through its own interface with the document and the numbered answer as shared state and one three-way choice question per sentence, using a 4-bit quantised model because the 8-bit one does not fit in memory beside the operating system.

Specialised encoders (B4). MiniCheck-RoBERTa [9] and HHEM-2.1-open [30] are the strongest small verifiers available. They score each (document, sentence) pair independently and return one probability of support; long documents are split into chunks of about 400 words and the maximum over chunks is kept, as in MiniCheck. They have no three-way output, so they are compared on the binary decision only.

Published results. For LLM-AggreFact, the scores of larger systems from the public leaderboard, such as Qwen2.5-7B used zero-shot, are reported as given and not re-run.

The slower systems are evaluated on stratified subsamples of the test sets: Rizzo-Flow on 300 answers of T1 and 600 pairs of T3, the untrained 3B on 600 answers of T1, and the generative judges on the 200 answers of the latency protocol below. Every other system is evaluated on the complete test sets.

5.3 Metrics

Because the share of hallucinated sentences differs between the development set and T1 (Section 4.6), the primary metrics do not depend on the base rate. For the three-way decision the main metric is the balanced accuracy, that is the mean recall over the three classes, together with the recall of each class and the share of each predicted class, which reveals a model that collapses onto one answer. For the binary decision the metrics are the balanced accuracy at a threshold of 0.5, as in the LLM-AggreFact leaderboard, and the AUROC of $P(\text{supported})$, which does not depend on any threshold. The lenient policy is evaluated with the balanced accuracy and AUROC of $1 - P(\text{contradicted})$. Calibration is measured with the ECE defined in Section 3.4, the Brier score and the negative log-likelihood, and selective prediction with risk–coverage curves and the area under them (AURC) [31]. All metrics on T1 are also reported per task and per band of K.

5.4 Latency protocol and hardware

Latency is measured on 200 answers of T1, 50 from each band of K, so that long answers are as well represented as short ones. Every system runs in process on MLX, one answer at a time, as a verifier would in a local assistant. The first three answers serve as warm-up and are excluded. For the System One variants each answer is timed three times and the median is kept; the generative judges are timed once, since greedy decoding is deterministic and each run is much longer. The reported time is the wall-clock time of the model call for one answer, from which the time per decision follows. Only one model runs at a time, and a memory guard runs every long job and records the free memory and the swap throughout. Results are written once and never overwritten, and each run stores its configuration, so every number in Chapter 6 can be traced to the run that produced it.

Chapter 6

Results

This chapter reports how GroundOne learns, what it achieves on the three frozen test sets, how it compares with the baselines, how well its probabilities are calibrated and how its latency grows with the number of decisions. It ends with an analysis of the errors that remain.

6.1 Training

Both models learned the task. Before training, a model reading the marker format has no notion of what the marker asks and its verdicts are at chance: the 0.5B answers "contradicted" for 88 % of the development sentences and reaches a three-way balanced accuracy of 0.32. In both runs the training loss fell below the value of a model that learned only the class frequencies ($\ln 3 \approx 1.10$) within the first few dozen steps, and the development score then rose steadily (Figure 6.1). The 0.5B, trained on all 13,203 answers, reached its best score at step 600 of 802, with a balanced accuracy of 0.633 averaged over the three tasks, after which it stayed flat. The 3B, trained for 12.25 hours on 6,500 answers, reached 0.705 at step 320 of 403 on its development subset. Measured on the full development set, the three-way balanced accuracy is 0.731 for the 0.5B and 0.784 for the 3B. All three classes were predicted throughout both runs, so neither model collapsed onto a single answer.

6.2 Accuracy on the RAG test set (T1)

On the 17,284 sentences of T1 the 0.5B model reaches a three-way balanced accuracy of 0.769 and the 3B model 0.832 (Table 6.1). On the binary decision the 0.5B reaches a balanced accuracy of 0.815 and an AUROC of 0.905, and the 3B 0.874 and 0.940. Both are far above the cues that never verify a sentence against

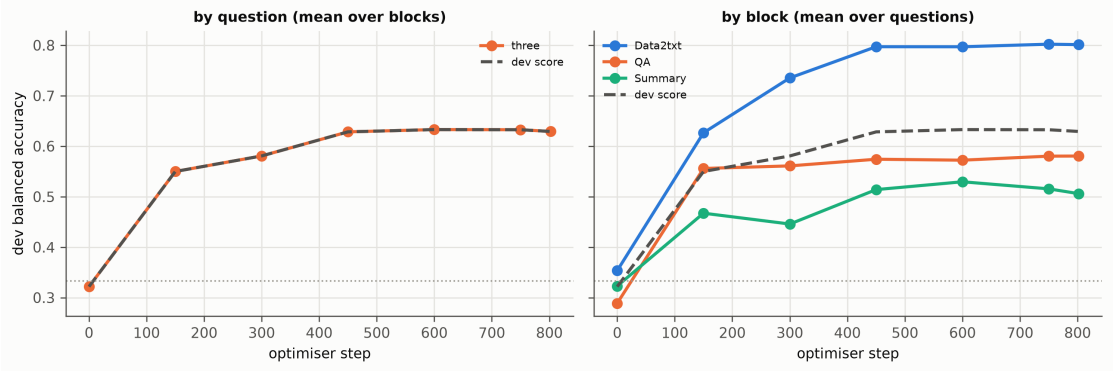


Figure 6.1: Three-way balanced accuracy on the development set during training of the 0.5B model, overall (left) and per task (right). The dotted line marks chance. The 3B shows the same shape on its 200-answer development subset (`figures/training-g1-3b-b1`).

its document: word overlap reaches an AUROC of 0.737, and the sentence-only classifier a three-way balanced accuracy of 0.570.

Table 6.1: GroundOne on T1, by task and by number of decisions per answer (K): three-way balanced accuracy, recall of contradicted sentences, and AUROC of P(supported).

		GroundOne 0.5B			GroundOne 3B		
Subset	Sentences	Bal. acc.	Contr.	AUROC	Bal. acc.	Contr.	AUROC
All	17,284	0.769	0.68	0.905	0.832	0.76	0.940
Question answering	4,876	0.589	0.03	0.916	0.699	0.30	0.948
Summarisation	4,538	0.544	0.13	0.775	0.629	0.30	0.873
Data-to-text	7,870	0.812	0.84	0.929	0.873	0.90	0.955
$K = 1$–4	2,434	0.606	0.18	0.857	0.677	0.25	0.921
$K = 5$–8	8,012	0.758	0.67	0.899	0.829	0.78	0.936
$K = 9$–13	5,201	0.824	0.82	0.926	0.863	0.86	0.945
$K \geq 14$	1,637	0.723	0.50	0.908	0.839	0.68	0.957

Three observations stand out. First, accuracy does not degrade with the length of the answer: answers with fourteen or more decisions are among the best handled, with 0.723 for the 0.5B and 0.839 for the 3B, and the hardest band is the shortest, which mostly contains question-answering and summarisation answers. This matters because long answers are where a System One saves the most time. Second, contradictions are recognised mainly in data-to-text, where RAGTruth concentrates them. In question answering and summarisation the 0.5B separates

supported from not-mentioned sentences well but finds almost no contradictions, with a recall of 0.03 and 0.13. The 3B raises both to 0.30, so model size compensates in part for the scarcity of such examples in the training data. Third, question answering and summarisation are the clearest evidence that the models read the document: there the sentence-only classifier stays at chance (0.36 and 0.35), while the two models reach 0.59–0.70 and 0.54–0.63.

6.3 Comparison with the baselines

Figure 6.2 compares all systems on T1. The specialised encoders reach an AUROC of 0.814 (MiniCheck-RoBERTa) and 0.821 (HHEM), above word overlap but well below both GroundOne models. The comparison by task qualifies this result: GroundOne's advantage comes mostly from data-to-text, where it is trained in domain and where the encoders are weakest (AUROC 0.71–0.75), while on summarisation MiniCheck-RoBERTa is better than the 0.5B model (0.872 against 0.775) and level with the 3B (0.873). The untrained models confirm that specialisation, not the readout mechanism, produces the accuracy: the 0.5B is at chance because it never answers "supported", and the 3B, asked the same question, reaches 0.437 against 0.818 after training on the same answers.

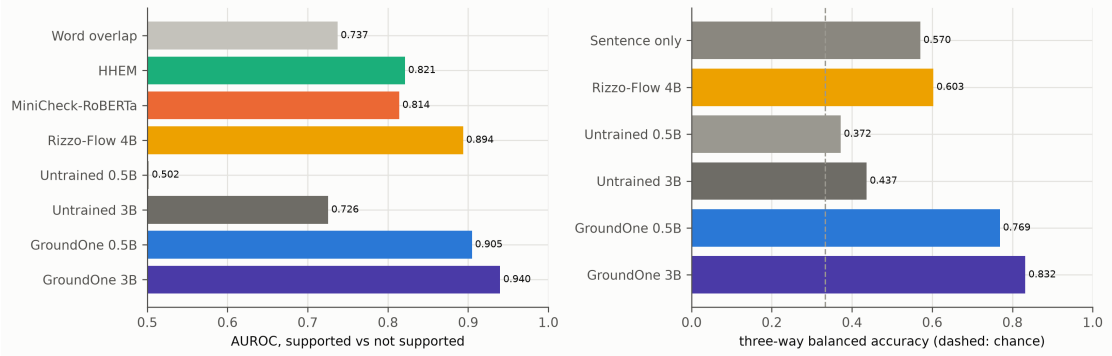


Figure 6.2: T1: AUROC of P(supported) for every system (left), and three-way balanced accuracy for the systems with a three-way output (right). Rizzo-Flow is evaluated on 300 answers and the untrained 3B on 600.

The general-purpose System One is the most instructive comparison. On the same 300 answers of T1, Rizzo-Flow, a 4B model fine-tuned for generic decisions, reaches a three-way balanced accuracy of 0.603, against 0.798 for the 0.5B GroundOne and 0.840 for the 3B; its AUROC (0.894) is close to that of the 0.5B (0.916) because it ranks sentences well but predicts "supported" for 90 % of them. A specialist eight times smaller therefore outperforms the generalist on the task it was built for.

On LLM-AggreFact (T2) the 0.5B model reaches a macro-averaged balanced accuracy of 70.1 over the eleven datasets, against 64.2 for the same model used zero-shot and 72.8 for the zero-shot 7B model on the public leaderboard. On the four datasets closest to retrieval-augmented generation the average is 74.4, equal to MiniCheck-RoBERTa (74.6); on the RAGTruth column it is 82.1 against 76.5, a comparison that favours GroundOne because it was trained on RAGTruth training answers. Outside retrieval-augmented generation GroundOne is weaker than the encoder by eight to nine points on AggreFact-CNN, AggreFact-XSum, FactCheck-GPT and REVEAL, the expected cost of training on RAG data only.

6.4 Transfer to a different domain (T3)

On VitaminC the picture reverses. The 0.5B reaches a three-way balanced accuracy of 0.442 and the 3B 0.527, with AUROCs of 0.722 and 0.817, while Rizzo-Flow reaches 0.712 and 0.909 on the same 600 pairs and MiniCheck-RoBERTa an AUROC of 0.968. Two facts put these numbers in context. MiniCheck-RoBERTa starts from the AlignScore model, whose training data include VitaminC [32], so T3 belongs to its training distribution, whereas for GroundOne it measures transfer from RAGTruth. Moreover, the gain of the 3B over the 0.5B on T3 comes from model size rather than from the RAG training: the untrained 3B already reaches 0.531 and an AUROC of 0.821. Specialisation on RAG answers therefore improves the RAG task substantially but does not, by itself, produce a general fact checker. Section 6.7 shows which cases are missed.

6.5 Calibration and selective prediction

The raw probabilities underestimate support. In the reliability diagram (Figure 6.3) the raw curve lies above the diagonal, because the class weights train the model as if the classes were balanced; the binary ECE on T1 is 0.30 for the 0.5B and 0.18 for the 3B, whose raw probabilities are already better calibrated. The temperature and bias fitted on the development set bring the ECE on T1 to 0.019 for both models and reduce the negative log-likelihood from 0.55 to 0.21 (0.5B) and from 0.36 to 0.17 (3B), while the AUROC is unchanged. The map fitted on development data therefore transfers to the test set of the same domain.

Calibration has a consequence that must be kept in mind. The bias restores the true frequency of each class, so the most probable class becomes “supported” more often: for the 0.5B on T1 the accuracy rises from 0.82 to 0.93, but the three-way balanced accuracy falls from 0.77 to 0.63. Decisions and probabilities therefore serve different purposes. When the three kinds of error matter equally, decisions are taken on the raw output or with a threshold chosen on development

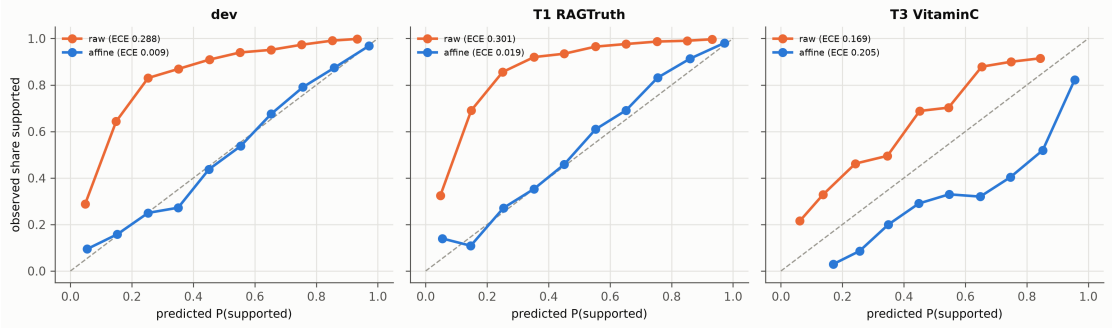


Figure 6.3: Reliability of P(supported) of the 0.5B before (raw) and after calibration on the development set, for the development set, T1 and T3. Points are bins with at least 20 sentences.

data; when a probability is used as a confidence, for a policy threshold or to decide which sentences to defer, the calibrated value is the right one. Used in this way, calibrated probabilities make selective prediction effective: deferring the least confident sentences lowers the error rate of the binary decision quickly, with an area under the risk–coverage curve of 0.015 (0.5B) and 0.010 (3B) on T1 (Figure 6.4).

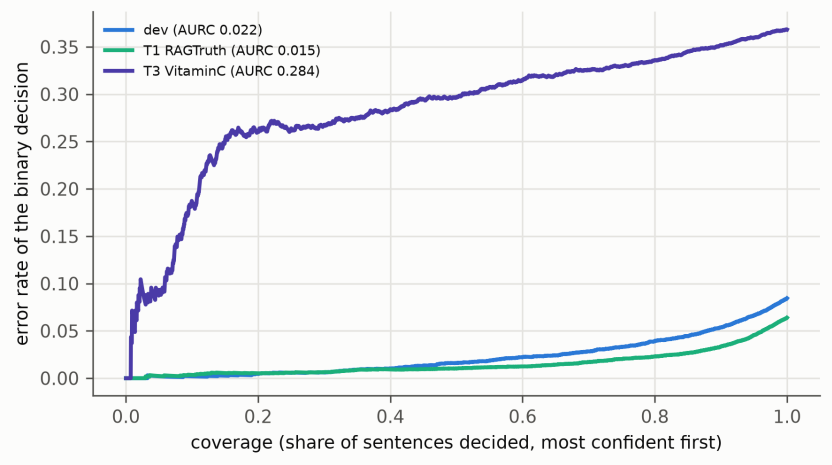


Figure 6.4: Risk–coverage curves of the binary decision of the 0.5B with calibrated probabilities: error rate as the least confident sentences are deferred.

Calibration does not transfer to VitaminC, where the binary ECE rises from 0.17 to 0.21 for the 0.5B and from 0.14 to 0.19 for the 3B. The bias encodes the base rate of RAGTruth, where about 88 % of sentences are supported, while VitaminC is balanced by construction. A calibration is valid for the distribution it was fitted on, so a deployment needs a small labelled sample of its own data.

6.6 Latency as a function of K

Figure 6.5 shows the central measurement of the thesis: the time needed to verify one answer, on the same 200 answers of T1, as the number of decisions grows. GroundOne is the fastest system at every size and in every band of K, and its advantage grows with K. For answers with fourteen or more decisions, the 0.5B GroundOne needs 0.41 seconds, against 0.80 for the terse judge and 2.51 for the verbose judge of the same size, which is 2.0 and 6.1 times faster; the 3B GroundOne needs 2.57 seconds, against 4.44 and 15.86, which is 1.7 and 6.2 times faster. Reading all verdicts in one pass is also faster than asking one question per sentence in a batch, by a factor of 1.4–1.5 at the highest K. Model size is the other lever: the 0.5B GroundOne verifies an answer about six times faster than the 3B (0.33 against 1.99 seconds at the median).

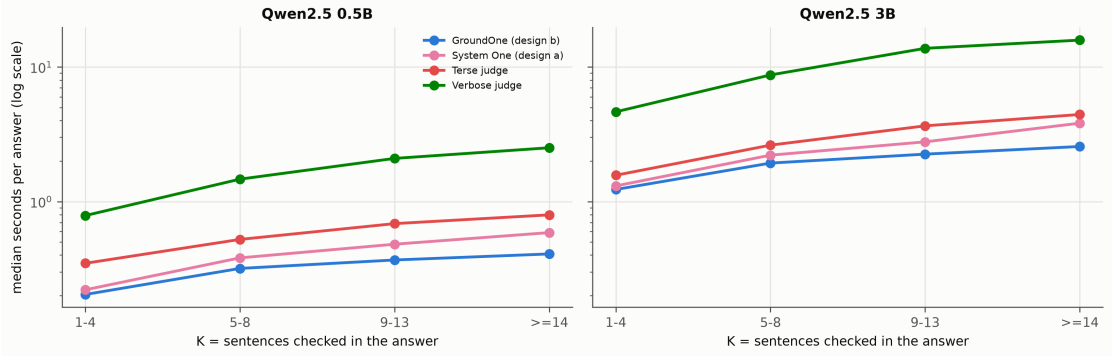


Figure 6.5: Median time per answer as a function of K on the same 200 T1 answers (in-process MLX, one answer at a time, M4 with 16 GB), for GroundOne, the System One with one question per sentence, and the terse and verbose generative judges. Logarithmic scale.

Speed is not the only difference (Figure 6.6). The untrained generative judges are also much less accurate and do not always respect the output format. The 0.5B verbose judge produces a parsable verdict for only 3 % of the sentences, because it mostly copies the answer back, and the 0.5B terse judge fails for 23 %. The 3B judges respect the format far better (0.1 % and 7.6 % failures) but reach a three-way balanced accuracy of only about 0.40, against 0.90 for the 3B GroundOne on the same answers. The verbose 3B judge writes on average 248 tokens per answer, all of which must be generated before the verdict can be used, while GroundOne generates none.

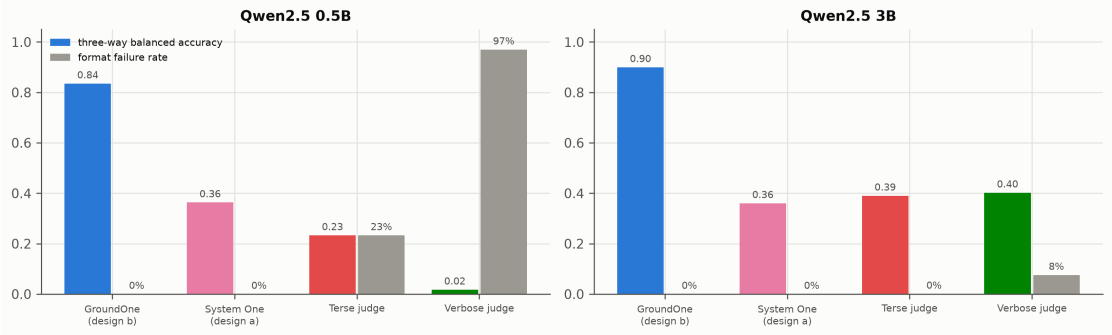


Figure 6.6: On the latency subset: three-way balanced accuracy (format failures counted as errors) and share of sentences whose verdict could not be parsed.

6.7 Error analysis

The weak result on VitaminC and the missed contradictions on T1 have a common cause, which word overlap reveals. On VitaminC, when a supported claim shares few words with its evidence (overlap below 0.4) the 0.5B recognises it in only 2 % of cases, against 64 % when the overlap exceeds 0.7, and it never recognises a contradiction in a claim without digits. The models rely partly on lexical overlap: a paraphrased claim looks "not mentioned", and a small change to a copied sentence, often a number, goes unnoticed.

The same weakness exists on T1, with less weight. Where a sentence shares few or some words with the document, the three-way balanced accuracy of the 0.5B is 0.77, but for sentences that copy more than 70 % of their words from the document it falls to 0.54, with a recall of 0.24 for contradicted and 0.43 for not-mentioned sentences. The 3B reduces this blind spot without removing it: on the same sentences its recall is 0.36 and 0.56, and its balanced accuracy 0.63. Hallucinations hidden in sentences that otherwise copy the source remain the main limitation of both models. VitaminC is not harder in absolute terms: it consists mostly of such cases, paraphrased support and minimal edits, which RAGTruth contains in smaller numbers.

Chapter 7

Discussion, Limitations and Threats to Validity

7.1 When a specialist System One pays off

7.2 Limitations

7.3 Threats to validity

Chapter 8

Conclusion and Future Work

8.1 Summary of findings

8.2 Future work

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